LESSON 3.1

Understand Ratios

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write and describe a ratio that compares two quantities.

WHY THIS MATTERS Welcome to Chef Academy! Head Chef Reyes is teaching new students how to read recipes like a pro. Her sparkling cranberry-apple fruit drink calls for 3 cups of apple juice for every 2 cups of sparkling water. Before the students can mix the drink, they need to understand how these ingredient amounts compare to each other.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	3 to 5 ■■■ □□□□□
Comparison *Spanish: Comparación*	Looking at two or more amounts to see how they are related.	5 eggs vs. 3 eggs — which recipe uses more?
Part-to-part *Spanish: Parte a parte*	A ratio comparing one part of a group to another part.	apples to oranges
Part-to-whole *Spanish: Parte a todo*	A ratio comparing one part to the whole group.	apples to all the fruit
Colon notation *Spanish: Notación con dos puntos*	Writing a ratio with two dots between the numbers, like 3:2.	3 to 2 can be written as 3:2
Tape diagram *Spanish: Diagrama de cintas*	A bar cut into equal parts that shows how two amounts in a ratio compare, and how they make up the whole.	3 equal parts of apple juice beside 2 equal parts of sparkling water — 5 parts of fruit drink in all.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A shelf has 6 math books and 9 reading books. Write reading to math, then math to all books.

1. Reading first: 9 : 6.
2. All books: 6 + 9 = 15, so math to all is 6 : 15.
3. Divide both terms by 3: 6 : 15 = 2 : 5.

Answer: 9 : 6 and 2 : 5

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A bowl has 4 apples and 1 orange. What two amounts are we comparing?

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The ratio of _____ to _____ is _____ .
- This means for every _____ _____ , there are _____ _____ .
- The order matters because _____ .
- I simplified by dividing both terms by _____ .

Word bank ratio | term | compare | for every | part | whole | total | simplest form | divide | order | first | second

WATCH OUT!

A common mistake

A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks

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Understand Ratios

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

A recipe uses 4 cups of milk and 1 cup of cream. What is the ratio of milk to cream?

(A) 1:4 (B) 4:1

(C) 4:5 (D) 5:1

STEPS TO TRY
1. Re-read the question: it asks for milk to cream, in that exact order.
2. In a ratio, the amount named first goes first. Find the milk amount and the cream amount before you write anything.
3. Underline the two amounts: milk = 4 cups, cream = 1 cup. Write them in the same order the question names them.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Chef Reyes uses 6 strawberries and 2 bananas in a smoothie. What is the part-to-whole ratio of bananas to total fruit?

(A) 2:8
(B) 2:6
(C) 6:2
(D) 8:2

STEPS TO TRY

1. Notice the word 'total' — this is a part-to-WHOLE ratio, not banana-to-strawberry.
2. First find the total fruit by adding strawberries and bananas. Then compare bananas to that total.
3. Add 6 + 2 to get the total fruit. Then write the bananas' number first, followed by that total.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

In a class of 30 students, 18 are girls. What is the part-to-part ratio of boys to girls?

(A) 12:18
(B) 12:30
(C) 18:12
(D) 18:30

STEPS TO TRY

1. The problem tells you girls (18) and the whole class (30) — but it asks about BOYS to girls.
2. Find the number of boys first by subtracting the girls from the class total. Then write boys first, girls second.
3. Subtract 30 − 18 to find the boys. Put that number first in the ratio, then 18 for the girls.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A pet store has 5 dogs and 9 cats. What is the ratio of dogs to cats?

(A) 5:9
(B) 5:14
(C) 9:5
(D) 14:5

STEPS TO TRY

1. Re-read: the question asks for dogs to cats, in that order.
2. A ratio only uses the two amounts being compared — don't add them together. Match the order to the words.
3. Dogs are named first, so their count (5) starts the ratio. Then write the cats' count after the colon.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

SENTENCE STARTER A ratio compares ___ to ___ using words like ___ or symbols like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help